



Hypothesis Testing in Public Health

This course is part of [Biostatistics in Public Health Specialization](#)

 Instructor: [John McGready, PhD, MS](#)

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Already enrolled

23,567 already enrolled

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6 modules

Gain insight into a topic and learn the fundamentals.

4.8 ★

661 reviews

Beginner level

Recommended experience ⓘ

Flexible schedule

2 weeks at 10 hours a week
Learn at your own pace

What you'll learn

- ✓ Use statistical methods to analyze sampling distribution
- ✓ Estimate and interpret 95% confidence intervals for single samples
- ✓ Estimate and interpret 95% confidence intervals for two populations
- ✓ Estimate and interpret p values for hypothesis testing

Skills you'll gain

Statistical Hypothesis Testing Biostatistics Scientific Methods Sampling (Statistics) Quantitative Research

Statistical Inference Public Health Analysis Statistical Methods Epidemiology

Details to know

 **Shareable certificate**
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 **Assessments**
11 assignments

 **Taught in English**
[22 languages available](#)

 **95%**
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Build your subject-matter expertise

This course is part of the [Biostatistics in Public Health Specialization](#). When you enroll in this course, you'll also be enrolled in this Specialization.

- Learn new concepts from industry experts
- Gain a foundational understanding of a subject or tool
- Develop job-relevant skills with hands-on projects
- Earn a shareable career certificate



There are 6 modules in this course

Biostatistics is an essential skill for every public health researcher because it provides a set of precise methods for extracting meaningful conclusions from data. In this second course of the Biostatistics in Public Health Specialization, you'll learn to evaluate sample variability and apply statistical hypothesis testing methods. Along the way, you'll perform calculations and interpret real-world data from the published scientific literature. Topics include sample statistics, the central limit theorem, confidence intervals, hypothesis testing, and p values.

[Show less](#)

Sampling Distributions and Standard Errors

Module 1 • 3 hours to complete

[Module details](#) ^

Within module one, you will learn about sample statistics, sampling distribution, and the central limit theorem. You will have the opportunity to test your knowledge with a practice quiz and, then, apply what you learned to the graded quiz.

What's included

 6 videos  1 reading  2 assignments

Hide info about module content ^

 6 videos • Total 74 minutes

Introduction • 3 minutes

Sampling Distribution Definition • 9 minutes

Examples: Sampling Distribution for a Single Mean • 16 minutes

Examples: Sampling Distribution for a Single Proportion, Incidence Rate • 16 minutes

Estimating Sampling Distribution Characteristics from Single Samples of Data • 22 minutes

Additional Examples • 7 minutes

 1 reading • Total 10 minutes

Formative Quiz 1 Solutions • 10 minutes

 2 assignments • Total 105 minutes

Sampling Distributions and Standard Errors • 45 minutes

Practice Quiz • 60 minutes

Confidence Intervals for Single Population Parameters

Module 2 • 3 hours to complete

[Module details](#) ^

Module two builds upon previous materials to discuss confidence intervals, the need for ample sizes of data, and ways to get around the need for ample sizes of data. The practice quiz helps you prepare for the graded quiz.

What's included

 6 videos  1 reading  2 assignments

Hide info about module content ^

Instructor

Instructor ratings  4.8 ★ (180 ratings)



[John McGready, PhD, MS](#)
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Confidence Intervals for Population Proportions and Incidence Rates • 15 minutes

So What Is a Confidence Interval, and How Should It Be Interpreted? • 14 minutes

A Note on Confidence Intervals for Population Quantities Based on Small Samples • 15 minutes

Additional Examples • 9 minutes

 1 reading • Total 10 minutes

Formative Quiz 2 Solutions • 10 minutes

 2 assignments • Total 120 minutes

Quiz: Confidence Intervals for Single Population Parameters • 60 minutes

Practice Quiz: Confidence Intervals for Single Population Parameters • 60 minutes

Confidence Intervals for Population Comparison Measures

[Module details](#) ^

Module 3 • 4 hours to complete

Within module three, confidence intervals are discussed at length and ratios are discussed again. Aside from the lectures, you will also be completing a practice quiz and graded quiz.

What's included

 8 videos  1 reading  2 assignments

Hide info about module content ^

 8 videos • Total 121 minutes

Introduction • 2 minutes

Confidence Intervals for Population Comparison Measures • 15 minutes

Confidence Intervals for Comparing Means of Continuous Outcomes Between Two Populations • 28 minutes

Confidence Intervals for Binary Comparisons: Part 1, Difference in Proportions • 12 minutes

Confidence Intervals for Binary Comparisons: Part 2, Ratio of Proportions and Odds Ratios • 24 minutes

Confidence Intervals for Incidence Rate Ratios • 10 minutes

A Brief Note About Ratios (Again) • 11 minutes

Additional Examples • 18 minutes

 1 reading • Total 10 minutes

Solutions, Formative Assessment • 10 minutes

 2 assignments • Total 120 minutes

Confidence Intervals for Population Comparison Measures • 60 minutes

Practice Quiz: Confidence Intervals for Population Comparison Measures • 60 minutes

Two-Group Hypothesis Testing: The General Concept and Comparing Means

[Module details](#) ^

Module 4 • 3 hours to complete

Within module four, you will look at statistical hypothesis tests, confidence intervals, and p-value. There is a practice quiz to prepare you for the graded quiz.

What's included

 6 videos  1 reading  2 assignments

Hide info about module content ^

 6 videos • Total 95 minutes

Introduction • 2 minutes

Hypothesis Testing: The General Concept, and for Comparing Means Between Two Populations • 18 minutes

(Hypothesis Testing) Comparing Means Between Two Populations: The Paired Approach • 22 minutes

(Hypothesis Testing) Comparing Means Between Two Populations: The Unpaired Approach • 21 minutes

Debriefing on the p-Value, Part 1 • 20 minutes

Additional Examples • 13 minutes

 1 reading • Total 10 minutes

Solutions, Formative Assessment • 10 minutes

 2 assignments • Total 60 minutes

Two-Group Hypothesis Testing: The General Concept and Comparing Means, and Earlier Course Material • 60 minutes

Practice Quiz: Two-Group Hypothesis Testing: The General Concept and Comparing Means • 0 minutes

Hypothesis Testing (Comparing Proportions and Incidence Rates Between Two Populations) & Extended Hypothesis Testing

[Module details](#) ^

Module 5 • 4 hours to complete

What's included

 11 videos  2 assignments

Hide info about module content ^

 11 videos • Total 138 minutes

Introduction • 2 minutes

(Hypothesis Testing) Comparing Proportions Between Two Populations: The “z-Test” Approach • 13 minutes

(Hypothesis Testing) Comparing Proportions Between Two Populations: Chi-Square and Fisher's Exact Test Approaches • 23 minutes

Hypothesis Tests for Comparing Incidence Rates Between Two Populations • 21 minutes

Debriefing on the p-Value and Hypothesis Testing, Part 2 • 9 minutes

Additional Examples • 9 minutes

Introduction • 3 minutes

(Hypothesis Testing) Comparing Means Between More Than Two Populations: Analysis of Variance (ANOVA) • 19 minutes

(Hypothesis Testing) Comparing Proportions Between More Than Two Populations: Chi-Square Tests • 14 minutes

Hypothesis Testing) Comparing Survival Curves Between More Than Two Populations: Log-Rank Tests • 8 minutes

Several Issues to Consider • 19 minutes

 2 assignments • Total 120 minutes

Hypothesis Testing (Comparing Proportions and Incidence Rates Between Two Populations) & Extended Hypothesis Testing • 60 minutes

Practice Quiz • 60 minutes

injectable contraception. The two research teams have asked you to help them interpret previously published results in order to inform the planning of their own studies. If you've already taken the Summarization and Measurement course, then this scenario will be familiar.

What's included

1 reading 1 assignment

Hide info about module content ^

1 reading • Total 30 minutes

Biostatistical Consulting Project • 30 minutes

1 assignment • Total 60 minutes

Course Project Quiz • 60 minutes



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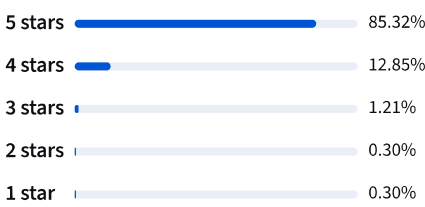
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Learner reviews

Showing 3 of 661

★ 4.8

661 reviews



A AB

★★★★☆ 4.0 • Reviewed on Jun 13, 2020

perfect except if there is a reference material, as PDFs, for self-revision after the course; no need to go back to the full video to remember everything

D DK

★★★★★ 5.0 • Reviewed on Jul 19, 2020

excellant descriptions, good examples and challenging practice sessions. Better if some more were added about ANOVA also. If it is considered as advanced , then it is ok. Good experience

K KC

★★★★★ 5.0 • Reviewed on Aug 24, 2021

The really great thing of this course is the professor! Outstanding! I just wish there were at least some recommended lectures/resources to calculate some of the exercises ourselves.

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